Gesture Recognition Fixed Action

This tutorial is based on the successful gesture recognition. You can view the [Gesture Recognition] tutorial by applying for a Baidu account. This section is based on gesture recognition having already successfully recognized gestures.

Main Code Content

Code path: /home/yahboom/Dofbot/6.AI_Visual/1.gesture_action.ipynb

```
#bgr8 to jpeg format
import enum
import cv2
def bgr8_to_jpeg(value, quality=75):
 return bytes(cv2.imencode('.jpg', value)[1])
```

```
#Import related modules
import threading
import time
from Arm_Lib import Arm_Device
# Create robotic arm object
Arm = Arm_Device()
time.sleep(.1)
```

```
import cv2
import time
import demjson
import pygame
from aip import AipBodyAnalysis
from aip import AipSpeech
from PIL import Image, ImageDraw, ImageFont
import numpy
import ipywidgets.widgets as widgets
# For specific gestures, please refer to the official documentation
https://ai.baidu.com/ai-doc/BODY/4k3cpywrv
hand={'One':'Number 1','Two':'Number 2','Three':'Number 3','Four':'Number 4',
 'Five':'Number 5', 'Six':'Number 6','Seven':'Number 7',
 'Eight':'Number 8','Nine':'Number 9','Fist':'Fist','Ok':'OK',
 'Prayer':'Prayer','Congratulation':'Worship','Honour':'Farewell',
 'Heart_single':'Heart gesture','Thumb_up':'Like','Thumb_down':'Dislike',
 'ILY':'I love you','Palm_up':'Palm up','Heart_1':'Double heart 1',
 'Heart_2':'Double heart 2','Heart_3':'Double heart 3','Rock':'Rock',
 'Insult':'Middle finger','Face':'Face'}
# The following key needs to be replaced with your own
""" Body Analysis APPID AK SK """
APP_ID = '18550528'
API_KEY = 'K6PWqtiUTKYK1fYaz13O8E3i'
SECRET_KEY = 'IDBUII1j6srF1XVNDX32I2WpuwBWczzK'
client = AipBodyAnalysis(APP_ID, API_KEY, SECRET_KEY)
```

```
g_camera = cv2.VideoCapture(0)
g_camera.set(3, 640)
g_camera.set(4, 480)
g_camera.set(5, 30) #Set frame rate
g_camera.set(cv2.CAP_PROP_FOURCC, cv2.VideoWriter.fourcc('M', 'J', 'P', 'G'))
g_camera.set(cv2.CAP_PROP_BRIGHTNESS, 40) #Set brightness -64 - 64 0.0
g_camera.set(cv2.CAP_PROP_CONTRAST, 50) #Set contrast -64 - 64 2.0
g_camera.set(cv2.CAP_PROP_EXPOSURE, 156) #Set exposure 1.0 - 5000 156.0
ret, frame = g_camera.read()
```

```
# Define camera display component
image_widget = widgets.Image(format='jpeg', width=600, height=500) #Set camera
display component
display(image_widget)
image_widget.value = bgr8_to_jpeg(frame)
```

```
#Define Chinese text conversion function
def cv2ImgAddText(img, text, left, top, textColor=(0, 255, 0), textSize=20):
 if (isinstance(img, numpy.ndarray)): # Check if it's OpenCV image type
 img = Image.fromarray(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
 # Create an object that can draw on the given image
 draw = ImageDraw.Draw(img)
 # Font style
 fontStyle = ImageFont.truetype(
 "simhei.ttf", textSize, encoding="utf-8")
 # Draw text
 draw.text((left, top), text, textColor, font=fontStyle)
 # Convert back to OpenCV format
 return cv2.cvtColor(numpy.asarray(img), cv2.COLOR_RGB2BGR)
```

```
#Define servo angles for different positions
look_at = [90, 164, 18, 0, 90, 90]
p_Prayer = [90, 90, 0, 180, 90, 180] #Prayer
p_Thumb_up = [90, 90, 90, 90, 90, 180] #Like
p_Heart_single = [90, 0, 180, 0, 90, 30] #Single heart gesture
p_Eight = [90, 180, 18, 0, 90, 90] #Eight
p_Congratulation = [90, 131, 52, 0, 90, 180] #Worship
p_Rock = [90, 0, 90, 180, 90, 0] #rock
p_fist = [90, 90, 0, 0, 90, 0] #fist
p_horse_1 = [90, 7, 153, 19, 0, 126] #
p_horse_2 = [90, 5, 176, 0, 0, 180]
p_horse_3 = [90, 62, 158, 0, 0, 0]
global running
running = 0
```

```
#Define function to move robotic arm, control servos 1-6 at the same time, p=
[S1,S2,S3,S4,S5,S6]
def arm_move_6(p, s_time = 500):
 for i in range(6):
 id = i + 1
 Arm.Arm_serial_servo_write(id, p[i], s_time)
 time.sleep(.01)
 time.sleep(s_time/1000)
```

```
# Define horse running
def horse_running():
 Arm.Arm_serial_servo_write(6, 150, 300)
 time.sleep(.3)
 Arm.Arm_serial_servo_write(6, 180, 300)
 time.sleep(.3)
```

```
global g_state_arm
g_state_arm = 0
def ctrl_arm_move(index):
 global running
 if index == "Prayer":
 arm_move_6(p_Prayer, 1000)
 time.sleep(1.5)
 arm_move_6(look_at, 1000)
 time.sleep(1)
 elif index == "Thumb_up":
 s_time = 500
 Arm.Arm_serial_servo_write(6, 180, s_time)
 time.sleep(s_time/1000)
 Arm.Arm_serial_servo_write(6, 90, s_time)
 time.sleep(s_time/1000)
 Arm.Arm_serial_servo_write(6, 180, s_time)
 time.sleep(s_time/1000)
 Arm.Arm_serial_servo_write(6, 90, s_time)
 time.sleep(s_time/1000)
 elif index == "Ok":
 s_time = 300
 Arm.Arm_serial_servo_write(4, 10, s_time)
 time.sleep(s_time/1000)
 Arm.Arm_serial_servo_write(4, 0, s_time)
 time.sleep(s_time/1000)
 Arm.Arm_serial_servo_write(4, 10, s_time)
 time.sleep(s_time/1000)
 Arm.Arm_serial_servo_write(4, 0, s_time)
 time.sleep(s_time/1000)
 elif index == "Heart_single":
 arm_move_6([90, 90, 90, 90, 90, 90], 800)
 time.sleep(.1)
 arm_move_6(p_Heart_single, 1000)
 time.sleep(1)
 elif index == "Five":
 arm_move_6(look_at, 1000)
 time.sleep(.5)
 elif index == "Eight":
 s_time = 300
 arm_move_6(p_Eight, 0)
 time.sleep(1)
 Arm.Arm_serial_servo_write(2, 165, s_time)
 time.sleep(s_time/1000)
 elif index == "Rock": #rock
 Arm.Arm_serial_servo_write6_array(p_Rock, 1300)
 time.sleep(3)
 Arm.Arm_serial_servo_write6_array(look_at, 1000)
 time.sleep(1)
 elif index == "Thumb_down": #Thumb down
 Arm.Arm_serial_servo_write6_array(p_horse_1, 1300)
```

```
 time.sleep(1)
 elif index == "Congratulation": #Worship
 Arm.Arm_serial_servo_write6_array(p_horse_2, 1000)
 time.sleep(1)
 running = 1
 while running == 1:
 horse_running()
 elif index == "Seven": #Number 7
 Arm.Arm_Buzzer_On(8) #Buzzer automatically sounds for 0.5 seconds
 Arm.Arm_serial_servo_write6_array(p_horse_3, 1000)
 time.sleep(2)
 Arm.Arm_serial_servo_write6_array(look_at, 1000)
 time.sleep(1)

 global g_state_arm
 g_state_arm = 0
```

```
#Move robotic arm to position where camera looks forward
arm_move_6(look_at, 1000)
time.sleep(1)
```

```
def start_move_arm(index):
 # Start robotic arm control thread
 global g_state_arm
 global running
 if g_state_arm == 0:
 closeTid = threading.Thread(target = ctrl_arm_move, args = [index])
 closeTid.setDaemon(True)
 closeTid.start()
 g_state_arm = 1

 if running == 1 and index == "Seven":
 running = 0
```

```
# Main process
try:
     Arm.Arm_Buzzer_On(1)
     s_time = 300
     Arm.Arm_serial_servo_write(4, 10, s_time)
     time.sleep(s_time/1000)
     Arm.Arm_serial_servo_write(4, 0, s_time)
     time.sleep(s_time/1000)
     Arm.Arm_serial_servo_write(4, 10, s_time)
     time.sleep(s_time/1000)
     Arm.Arm_serial_servo_write(4, 0, s_time)
     time.sleep(s_time/1000)

     while True:
         """1.Take photo """
         ret, frame = g_camera.read()
         #image = get_file_content('./image.jpg')
         """ 2.Call gesture recognition """
         raw = str(client.gesture(image_widget.value))
         text = demjson.decode(raw)
         try:
```

```
        res = text['result'][0]['classname']
        except:
            # print('Recognition result: Nothing detected~' )
            # img = cv2ImgAddText(frame, "Not recognized", 250, 30, (0, 0 ,
255), 30)
            img = frame
        else:
            # print('Recognition result:' + hand[res])
            # img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)
           www.yahboom.com
        if res == 'Prayer': # 1 Prayer
            print('Recognition result:' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)
            start_move_arm(res)
        elif res == 'Thumb_up':# 2 Like
            print('Recognition result:' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)
            start_move_arm(res)
        elif res == 'Ok': # 3 OK
            print('Recognition result:' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)
            start_move_arm(res)
        elif res == 'Heart_single': # 4 Single heart gesture
            print('Recognition result:' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)
            start_move_arm(res)
        elif res == 'Five': # 5 Number 5
            print('Recognition result:' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)
            start_move_arm(res)
        elif res == "Eight": # Number 8
            print('Recognition result:' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)
            start_move_arm(res)

        elif res == "Rock": # rock
            print('Recognition result:' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)
            start_move_arm(res)
        elif res == "Congratulation": # Worship
            print('Recognition result:' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)
            start_move_arm(res)
        elif res == "Seven": # Number 7
            print('Recognition result:' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)
            start_move_arm(res)
        elif res == "Thumb_down": # Thumb down
            print('Recognition result:' + hand[res])
            img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)
            start_move_arm(res)

        else:
           img = frame
 image_widget.value = bgr8_to_jpeg(img)
except KeyboardInterrupt:
     print(" Program closed! ")
     pass
```

If the set gesture is recognized, the robotic arm will perform the corresponding action. Currently, the gesture and action correspondence is as follows:

Prayer	**The robotic arm prays**
Thumb_up	The robotic arm's claws mimic clapping
Ok	The robotic arm nods
Heart_single (One-handed heart shape)	The robotic arm kneels in greeting
Number Five	The robotic arm returns to its initial position
Number Eight	The robotic arm sees a gun and reacts with fright
Rock	The robotic arm falls forward
Congratulation	The owner mounts the horse and starts running
Number Seven	The horse pulls back from the brink and stops running
Thumb_down	The robotic arm assumes a horse-like posture while waiting for the owner to get in the car